Software Development Proposal

Daikibo Factory Telemetry Dashboard

Overview

This proposal outlines the development of a private web-based telemetry dashboard for Daikibo's manufacturing facilities. The proposed system will provide a centralized interface for monitoring the health status of machines installed across four factories. Each factory contains nine monitored machines that continuously send telemetry data.

The dashboard will allow authorized employees to securely access real-time machine health information through the company intranet. Authentication will be integrated with Daikibo's internal authentication server, enabling employees to use their existing company credentials.

The application will feature a single-page dashboard with expandable and collapsible views for factories and individual devices. Users will also be able to review the recent history of machine statuses for monitoring and troubleshooting purposes.

The proposed solution focuses on reliability, usability, security, and ease of maintenance.

Scope

The scope of this project includes the design, development, testing, and deployment of a secure telemetry monitoring dashboard.

Functional Requirements

1. Secure Internal Access

- The application will only be accessible within Daikibo's internal network (intranet).
- External/public access will be restricted.
- User authentication will be synchronized with the company's internal authentication server.
- Employees will log in using their existing company-wide accounts.

2. Telemetry Dashboard

- A single-page dashboard will display the health status of all monitored machines.
- The dashboard will support monitoring for:
    - 4 factories
    - 9 machines per factory
    - Total: 36 monitored devices

3. Factory-Level Expand/Collapse View

- Users can expand or collapse each factory section.
- The collapsed view will provide a quick summary of machine statuses.

- The expanded view will show detailed information for all devices in the selected factory.

4. Device-Level Expand/Collapse View

- Each machine entry can be expanded to display additional details.
- Historical telemetry statuses will be displayed for troubleshooting and monitoring purposes.
- Users can review recent operational history for each device.

5. Status Indicators

The dashboard will visually represent machine health using status indicators such as:

- Healthy
- Warning
- Offline
- Error/Critical

6. Responsive User Interface

- The interface will be optimized for desktop usage.
- The application will support standard modern web browsers used within the organization.

Graphics and Interface Reference

The dashboard interface will follow the graphics and layout structure provided in the supplied proposal template. The UI will prioritize:

- Clear factory grouping

- Easy device navigation
- Readable telemetry history
- Simple expandable/collapsible interactions
- Professional industrial dashboard appearance

Estimate

The estimated effort for the project is approximately 140 man-hours.

Task	**Estimated Hours**
Requirements Analysis & Planning	12 Hours
UI/UX Design	16 Hours
Frontend Dashboard Development	42 Hours
Backend API Integration	24 Hours
Authentication Integration	14 Hours
Telemetry Data Handling	10 Hours
Testing & Quality Assurance	14 Hours
Deployment & Integration	8 Hours
Documentation & Handover	10 Hours
Total	140 Hours

Resource Allocation

- Frontend Developer
- Backend Developer

- QA/Test Engineer
- Project Coordinator

Timeline

The estimated project duration is 5 weeks.

Milestone	Duration
Project Kickoff & Requirement Confirmation	Week 1
UI/UX Design Completion	Week 1
Frontend Development	Week 2
Backend & Authentication Integration	Week 3
Telemetry Integration & Dashboard Completion	Week 4
Testing, Bug Fixes & Deployment	Week 5

Key Deliverables

- Secure intranet-based dashboard
- Authentication integration
- Expandable/collapsible telemetry interface
- Device status monitoring
- Historical telemetry display
- Deployment documentation

Support

Post-deployment support will be provided to ensure the reliability and long-term success of the dashboard solution.

Included Support Services

- Bug fixes and issue resolution
- Technical support tickets
- Performance monitoring assistance
- Minor UI improvements
- Security updates and maintenance
- Assistance with future feature enhancements

Additional feature requests and extended maintenance services can be arranged under a separate support agreement if required.

Conclusion

The proposed telemetry dashboard solution will provide Daikibo with a secure and centralized platform for monitoring machine health across all factories. The system will improve operational visibility, simplify troubleshooting, and provide employees with fast access to telemetry information through an intuitive interface.

We appreciate the opportunity to submit this proposal and look forward to collaborating with Daikibo on this project.